EAST WEST UNIVERSITY

Department of Mathematical and Physical Sciences

LABORATORY REPORT

CHE109- ENGINEERING CHEMISTRY

Experiment No: 05

EXPERIMENT TITLE

Standardization of $Na_2S_2O_3$ Solution with a Standard $K_2Cr_2O_7$ Solution

SUBMITTED TO

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Experiment 5: Standardization of $Na_2S_2O_3$ Solution with a Standard $K_2Cr_2O_7$ Solution

Objective

The strength (molarity) of the supplied sodium thiosulfate ($Na_2S_2O_3$) solution was determined by standardizing it against a 0.010 M potassium dichromate ($K_2Cr_2O_7$) solution.

Introduction and Principle

In acidic medium, potassium dichromate oxidizes iodide ions to liberate iodine. The liberated iodine is then titrated with sodium thiosulfate, which is oxidized to tetrathionate. The amount of iodine formed is stoichiometrically equivalent to the dichromate present. Starch is used near the endpoint to make the disappearance of iodine easier to observe.

Reactions

$K_2Cr_2O_7(aq) + 14HCl(aq) + 6KI(aq) \rightarrow 2CrCl_3(aq) + 3I_2(aq) + 8KCl(aq) + 7H_2O(l)$ (in the dark)

$I_2 + I^- \leftrightarrow I_3^-$(Triiodide)

$6Na_2S_2O_3(aq) + 3\ I_2[I_3^-]_{(g)} \rightarrow 6NaI(aq) + 3Na_2S_4O_6(aq)$ (during titration)

(Starch + I_3^- ↔ starch–I_3)

Chemicals*:* $K_2Cr_2O_7$, $Na_2S_2O_3$, $NaHCO_3$, concentrated HCl, 10% KI solution, starch solution, and distilled water.

Apparatus: Conical flask, volumetric flask, burette, dropper, funnel, pipette filler, watch glass, and other usual laboratory glassware.

Procedure

a) Preparation of 0.01 M 250 mL $K_2Cr_2O_7$ solution

A mass of 0.735 g of $K_2Cr_2O_7$ was weighed using W = SMV/1000, where the molar mass of $K_2Cr_2O_7$ is 294 g/mol.
The weighed $K_2Cr_2O_7$ was transferred to a 250 mL volumetric flask, and a small amount of distilled water was added.
The flask was shaken until the solute dissolved, and distilled water was added up to the 250 mL mark.

b) Standardization of $Na_2S_2O_3$ solution with standard $K_2Cr_2O_7$ solution

1. The supplied $Na_2S_2O_3$ solution was placed in the burette, and the initial burette reading was recorded.
2. A 50 mL portion of distilled water was measured into a conical flask.

3. One gram of $NaHCO_3$ was added, and the flask was shaken until the salt dissolved.
4. Three millilitres of concentrated HCl was added slowly while the flask was gently rotated to mix the liquid.
5. Ten millilitres of $K_2Cr_2O_7$ solution was added to the conical flask, and the solution was mixed well.

6. Four milliliters of 10% KI solution was added to the flask, and the contents were mixed well. The flask was immediately covered with a watch glass and left in the dark for about five minutes; the solution became dark brown.

7. The flask was removed from the dark area.

8. The liberated iodine was titrated with $Na_2S_2O_3$ from the burette until the colour faded to pale yellow. Ten drops of starch solution were added; the solution turned deep violet.

9. Sodium thiosulfate was added continuously until the solution turned light green or light blue. The final burette reading was recorded as the endpoint.

10. The titration steps were repeated at least two more times, and the average volume of $Na_2S_2O_3$ used was calculated.

Experimental Data

Standardization of $Na_2S_2O_3$ solution with standard 0.01 M $K_2Cr_2O_7$ solution.

Observation	Initial burette reading (mL)	Final burette reading (mL)	$Na_2S_2O_3$ used (mL)
1	4.0	16.4	12.4
2	18.5	30.7	12.2
3	33.2	45.3	12.1
Average			12.23

Calculations

From the reaction stoichiometry: 1 mol $K_2Cr_2O_7$ = 3 mol I_2 = 6 mol $Na_2S_2O_3$.

$$M(Na_2S_2O_3) = \frac{6 \times M(K_2Cr_2O_7) \times V(K_2Cr_2O_7)}{V(Na_2S_2O_3)} = \frac{6 \times 0.01 \times 10}{12.23}$$

$$= 0.04906 \text{ M}$$

$$\approx 0.0491 \text{ M}.$$

Percentage Error

$$\% \text{ error} = \left(\left| \frac{|\text{theoretical value} - \text{experimental value}|}{theoritical\ value} \right| \right) \times 100$$

$$= \left(\left| \frac{0.06 - 0.0491}{0.06} \right| \right) \times 100$$

$$= 18.17\%$$

Result

The strength of the supplied $Na_2S_2O_3$ solution was found to be approximately 0.0491 M, based on an average titre of 12.23 mL.

Discussion / Error Analysis

The three titre volumes (12.4, 12.2, and 12.1 mL) were close to one another. The calculated concentration was 0.0491 M, giving an 18.17% error relative to the 0.0600 M reference value. Differences may have resulted from endpoint judgement, burette reading, or solution-preparation and transfer measurements. The iodine mixture was kept in the dark as directed, and starch was added near the endpoint so the colour change could be observed clearly.

Answers to Practice Questions

1. **Why was $NaHCO_3$ added?**

A $NaHCO_3$ was added to neutralize excess acid and reduce the acidity of the medium, helping to minimize decomposition of thiosulfate by strong acid.

2. **Why was the flask kept in a dark area during iodometry?**

A The flask was kept in the dark to limit light-induced side reactions and loss of iodine, helping the liberated iodine remain available for titration.

3. **Why was potassium iodide used instead of iodine solution?**

A KI was used because iodide ions are oxidized by dichromate in acidic solution to produce an equivalent amount of iodine in the flask. This generates iodine in situ from the known amount of primary-standard dichromate.

4. **What is meant by iodometry?**

A Iodometry is an indirect redox titration in which an oxidizing agent liberates iodine from iodide, and the liberated iodine is titrated with a standard reducing solution such as sodium thiosulfate.

5. **Why was starch added near the end of the titration?**

A Starch forms an intense blue or violet complex with iodine/triiodide, making small amounts of iodine easy to detect. It was added near the endpoint to avoid a strongly bound complex that could make the endpoint slow or less accurate.

6. **Discuss the precautions taken during the experiment.**

A The flask was covered with a watch glass and kept in the dark for about five minutes after KI was added. The burette reading was taken at eye level, the flask was mixed while titrant was added, starch was added only after the solution became pale yellow, and sodium thiosulfate was added carefully near the endpoint until the colour became light green or light blue.